

Extended Mean-Field Analysis of the Stacked-Triangular Ising Antiferromagnet

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Phase transitions of the frustrated Ising model on a stacked-triangular lattice with the antiferromagnetic nearest-neighbor and the ferromagnetic next-nearest-neighbor interactions are studied. For this system, an earlier mean-field analysis yielded successive phase transitions, with a partial disordered phase stabilized in an intermediate temperature range and ferrimagnetic phases stabilized at lower temperatures. We have performed an extended mean-field analysis based on the variational principle in which correlation effects in the frustrated triangular layer are taken into account. It is found that the partial disordered phase is largely suppressed relative to the ferrimagnetic phase due to the correlation effects.

[triangular lattice antiferromagnet, frustration, ferrimagnetic phase, partial disordered phase, mean-field approximation]

§1. Introduction

In these two decades, effects of spin frustration on the nature of various magnetic phase transitions have been studied both experimentally and theoretically. Hexagonal antiferromagnets CsCoCl_3 and CsCoBr_3 are typical examples of such frustrated magnets in which magnetic ions (Co^{2+}) form a stacked-triangular (or simple hexagonal) lattice and are coupled via the antiferromagnetic exchange interactions with easy-axis-type anisotropy. The antiferromagnetic interactions in the triangular layer give rise to geometry-induced frustration. It has been known for some time that phase transitions of these compounds exhibit many unusual features, apparently due to the intrinsic frustration effect.^{1–7)} Since Co^{2+} has spin $1/2$ and the exchange interaction between Co^{2+} has fairly strong easy-axis-type anisotropy, the standard Ising model is expected to offer a reasonable description of these materials.

Phase transitions of the frustrated Ising model on the two-dimensional triangular lattice were theoretically studied by Mekata in conjunction with the magnetic phase transition of CsCoCl_3 and CsCoBr_3 .⁸⁾ Applying the mean-field approximation to the Ising model

on the triangular lattice with the antiferromagnetic nearest-neighbor interaction ($J_1 < 0$) and the ferromagnetic next-nearest-neighbor interaction ($J_2 > 0$), Mekata found three distinct ordered phases and successive phase transitions among them. These three ordered phases were called a partial disordered phase, a three-sublattice ferrimagnetic phase and a two-sublattice ferrimagnetic phase.

A triangular lattice consists of three sublattices. In the partial disordered phase, one of these three sublattices is disordered, while the other two are ordered in mutually opposite directions with the same magnitude of the sublattice magnetization. This state may be denoted as $(m, 0, -m)$; see Fig. 1(a). In the two-sublattice ferrimagnetic phase, the sublattice magnetizations of the two sublattices are equal while that of the other sublattice points oppositely; see Fig. 1(b). This state may be denoted as $(m, m, -m)$. In the three-sublattice ferrimagnetic phase, all the sublattice magnetizations are mutually different and the ordered state does not have reflection or inversion symmetry; see Fig. 1(c).

It was found in ref. 7 that, when the ratio $R \equiv J_2/|J_1|$ is smaller than a critical value $R_m \approx 0.4$, the system exhibits *three* successive transitions on decreasing temperature; first

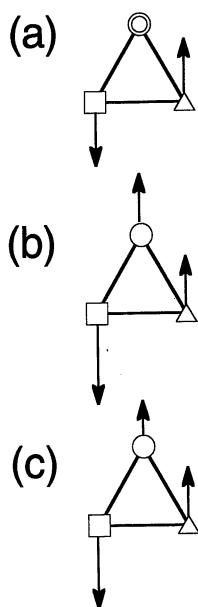


Fig. 1. Ordered-state spin configurations in the triangular-lattice Ising antiferromagnet; (a) a partial disordered phase, (b) a three-sublattice ferrimagnetic phase and (c) a two-sublattice ferrimagnetic phase. Symbols $\circ$, $\triangle$, $\square$ denote three sublattices of the original triangular lattice.

from the paramagnetic phase to the partial disordered phase at $T=T_{N1}$, then from the partial disordered phase to the three-sublattice ferrimagnetic phase at $T=T_{N2}$, and finally from the three-sublattice ferrimagnetic phase to the two-sublattice ferrimagnetic phase at $T=T_{N3}$. At $R=R_m$, these three transition temperatures T_{N1} , T_{N2} and T_{N3} merge yielding a single multicritical point. Above R_m , the system exhibits only one phase transition from the paramagnetic phase to the two-sublattice ferrimagnetic phase. Phase transitions from one state to the other are all continuous.

In particular, exotic ordered phases such as the partial disordered phase and the three-sublattice ferrimagnetic phase were found to be stabilized over a wide temperature range. Mekata suggested that these exotic phases found by the mean-field analysis might indeed be realized in CsCoCl_3 and CsCoBr_3 .⁸⁾ The occurrence of three successive transitions in CsCoCl_3 similar to those found by Mekata were also suggested by Plumer, Caillé and Hood based on a Landau-type free-energy analysis,⁹⁾ and by Kaburagi, Tonegawa and Kanamori based on a cluster variation

method.¹⁰⁾

One has to remember that, at the mean-field level, there is no real distinction between two- and three-dimensional systems. In fact, however, the transition behavior is quite different between in two and in three dimensions, and mean-field results are generally much better for three-dimensional systems than for two-dimensional systems. In the case of the purely *two-dimensional* triangular antiferromagnet with the nearest- and next-nearest-neighbor interactions, it now seems well established that there exist two 'Kosterlitz-Thouless-Type' transitions, with the intermediate phase characterized by the power-law-decaying spin correlations and with the low-temperature two-sublattice ferrimagnetic phase.¹¹⁾ Therefore, our interest in the present paper concerns with the *three-dimensional* stacked-triangular antiferromagnets.

Following the pioneering work by Mekata, several theoretical works have been made for the frustrated Ising antiferromagnet on the three-dimensional stacked-triangular lattice. On the basis of a Landau-Ginzburg-Wilson analysis and a Monte Carlo simulation, Blankschtein, Ma, Berker, Grest and Soukoulis¹²⁾ suggested the occurrence of two successive phase transitions with the intermediate partial disordered phase and the low-temperature two-sublattice ferrimagnetic phase. They claimed that the upper transition is continuous whereas the lower transition between the partial disordered phase and the two-sublattice ferrimagnetic phase is first order. Similar conclusion was also reported by Netz and Berker.¹³⁾

Matsubara and coworkers performed a Monte Carlo simulation on the three-dimensional stacked-triangular Ising antiferromagnet with the antiferromagnetic in-plane nearest-neighbor and the ferromagnetic in-plane next-nearest-neighbor interactions.¹⁴⁾ Although no definite conclusion was reported concerning the stability of the partial disordered phase by these authors, their data seem to suggest that the ordered phase always has a ferrimagnetic character in contrast to the mean-field prediction. More recently, several intensive Monte Carlo simulations have been made focusing on the critical properties at

$T = T_{N1}$.¹⁵⁾

Various experimental information is now available both for CsCoCl_3 and CsCoBr_3 . For CsCoCl_3 , neutron diffraction measurements were carried out by Melamud, Pinto, Nakovsky and Shaked.¹⁾ These authors determined the magnetic structure at 4.2 K to be a two-sublattice ferrimagnetic phase, and also suggested that the magnetic structure near T_{N1} differs from the one at 4.2 K. Mekata and Adachi further performed a neutron scattering study of CsCoCl_3 and suggested the occurrence of three successive transitions with the partial disordered phase stabilized near T_{N1} , which is consistent with the mean-field result. The occurrence of the lower transition at T_{N2} was also suggested from the NMR measurements by Ajiro *et al.*,⁴⁾ who observed a clear anomaly in the spin dynamics at T_{N2} .

For CsCoBr_3 , neutron diffraction measurements were made by Yelon, Cox and Einschütz, who claimed that the partial disordered phase is stabilized near T_{N1} while the ferrimagnetic phase becomes stable at lower temperatures.²⁾ More recently, Farkas, Gaulin, Tun and Briat suggested based on neutron diffraction measurements that there exist three successive transitions in CsCoBr_3 .⁵⁾

By contrast, in the specific-heat measurements of CsCoCl_3 and CsCoBr_3 , no discernible anomaly could be detected either around T_{N2} or T_{N3} , while a clear anomaly was observed at T_{N1} .^{2,7)} This observation casts some doubt to the view that T_{N2} and T_{N3} correspond to true thermodynamic phase transitions. Furthermore, in neutron scattering measurements of CsCoCl_3 by Endoh *et al.*, a weak (1, 1, 1) intensity characteristic of a ferrimagnetic phase has clearly been observed just below T_{N1} .⁶⁾ This observation suggests that the ferrimagnetic phase is always stabilized below T_{N1} and there is no partial disordered phase in the strict sense.

Thus, the stability of the partial disordered phase and the occurrence of true thermodynamic transitions at T_{N2} and T_{N3} still remain ambiguous both experimentally and theoretically. Mekata's theoretical analysis was based on the naive mean-field approximation which completely neglected correlation effects. Usually, a mean-field approximation is

known to give a qualitatively correct result for *three-dimensional* systems. However, in highly frustrated systems like the present model, it is not obvious whether the mean-field approximation could reproduce the transition behavior even qualitatively.

Shiba modified the naive mean-field approximation in such a way that the correlation effects associated with the interplane interactions along the *c*-axis were taken into account exactly, while the intraplane interaction was treated as mean field.¹⁶⁾ The resulting phase diagram is qualitatively similar to the one obtained by Mekata. In particular, the partial disordered phase is found to be stabilized over a rather wide temperature range. Since, in this approach, the competing intraplane interactions are dealt with as mean field as in the case of the naive mean-field approximation, and since frustration in the triangular layer is expected to be a key ingredient of the transition, it is not necessarily clear how the correlation effects in the frustrated triangular layer modify the results.

Our present study was motivated by the recent theoretical work by Nakanishi on a three-dimensional axial-next-nearest-neighbor Ising (ANNNI) model, in which the nearest-neighbor and next-nearest-neighbor antiferromagnetic interactions compete along an axial direction. For this model, an earlier mean-field analysis gave the results qualitatively similar to the Mekata's results for the triangular-lattice Ising model, including the occurrence of successive transitions and the existence of the partial disordered phase.¹⁷⁾ By contrast, by treating the interactions along the axial direction of the model exactly, Nakanishi found that neither the partial disordered nor the three-sublattice ferrimagnetic phase appears, at least in the case of period-three modulation.¹⁸⁾ This observation has lead us to reexamine the transition behavior of the stacked-triangular Ising antiferromagnet based on an extended mean-field approximation in which the correlation effects in the frustrated triangular layer are taken into account. More specifically, we deal with the intraplane interactions along the ladder shown as bold lines in Fig. 2 exactly as the one-dimensional Ising model, while the residual intraplane inter-

actions between the neighboring ladders as well as the interplane interactions along the c -axis are dealt with as mean field.

It has been found that the partial disordered phase still survives as an ordered state within our approximation. By contrast, as compared with the previous mean-field results, the partial disordered phase is found to be largely suppressed relative to the ferrimagnetic phase. This observation suggests that the correlation effects associated with the frustrated triangular layer enhance the two-sublattice ferrimagnetic phase and suppress the partial disordered phase.

This paper is organized as follows. In §2, after presenting the model, we explain our approximation and the method of calculation. The obtained results are presented in §3. Magnetic phase diagrams are also given. Section 4 is devoted to summary and discussion. Comparisons are made with the corresponding mean-field phase diagram obtained by Mekata.

§2. Model and Analysis

We start with the following Hamiltonian,

$$\mathcal{H} = -J_1 \sum_{\langle ij \rangle} \sigma_i \sigma_j - J_2 \sum_{\langle ij \rangle} \sigma_i \sigma_j - J_0 \sum_{\langle ij \rangle} \sigma_i \sigma_j, \quad (2.1)$$

where $\sigma_i = \pm 1$ denotes an Ising-spin variable sitting on the i -th site of the three-dimensional stacked-triangular lattice. The first and the second summations are taken over all nearest- and next-nearest-neighbor pairs on the triangular layer, while the third summation is taken over all interplane nearest-neighbor pairs along the c -axis.

We assume that the nearest-neighbor interaction in each triangular layer is antiferromagnetic while the next-nearest-neighbor interaction is ferromagnetic ($J_1 < 0$, $J_2 > 0$). In zero magnetic field, the sign of the interplane interaction J_0 is irrelevant since the model is classical. In real materials such as CsCoCl_3 and CsCoBr_3 , the interplane interaction J_0 is an-

tiferromagnetic. In the following, however, we assume the interplane interaction to be ferromagnetic, ($J_0 > 0$) just for simplicity of calculation.

In making comparison with the results of the previous calculations, some particular combinations of the coupling constants are of interest. Thus, we first investigate the 'two-dimensional' case studied by Mekata, where one has $J_0 = 0$. Then, we study the effects of the interplane interaction by setting $J_0 > 0$. The cases so far studied by Monte Carlo simulations mostly correspond to $J_0 = J_1$ and $J_2 = 0$ (or with small but finite $J_2 > 0$). Note, however, that in real materials CsCoCl_3 and CsCoBr_3 the interplane interaction J_0 is much bigger than the intraplane interactions; namely, one has $|J_0| \gg |J_1|, J_2$. In such a case, our following approximation might become poor, since it neglects the correlation effects associated with the interplane interaction along the c -axis.

Our extended mean-field analysis is based on the standard variational principle for the free energy

$$\mathcal{F} \leq \mathcal{F}_{\text{trial}} = \mathcal{F}_0 + \langle \mathcal{H} - \mathcal{H}_0 \rangle_0, \quad (2.2)$$

where $\mathcal{F}_0$ is the free energy of the trial Hamiltonian $\mathcal{H}_0$, and $\langle \cdots \rangle_0$ denotes a thermal average with respect to $\mathcal{H}_0$. In the standard mean-field approximation, one takes a noninteracting single-spin Hamiltonian as a trial Hamiltonian, $\mathcal{H}_0 = \sum_i H_i \sigma_i$, H_i being the variational parameter to be determined so as to minimize the r.h.s of eq. (2.2). In the present calculation, in order to take account of the correlation effects in the triangular layer, we introduce the one-dimensional Ising model as our trial Hamiltonian $\mathcal{H}_0$ which is treated exactly, while the residual interactions are treated as mean field. More specifically, the interactions along bold lines in the ladder shown Fig. 2 are treated exactly as the one-dimensional Ising model, while the interactions along thin lines in Fig. 2 as well as the interplane interactions are treated as mean field.

The trial Hamiltonian $\mathcal{H}_0$ may be written as

$$\mathcal{H}_0 = \sum_l H_{1D}^{(l)}, \quad (2.3)$$

$$\mathcal{H}_{1D}^{(l)} = -J_1 \sum_i \sigma_{li} \sigma_{li+1} - J_1 \sum_i \sigma_{li} \sigma_{li+2} - J_2 \sum_i \sigma_{li} \sigma_{li+3} - \sum_i H_{li} \sigma_{li}, \quad (2.4)$$

where $\mathcal{H}_{1D}^{(l)}$ is the Hamiltonian of the l -th chain (ladder) which has up to the third-nearest-neighbor interactions, while σ_{li} denotes the i -th Ising spin on the l -th chain. The variational parameter H_{li} should be determined so as to minimize the trial free energy $\mathcal{F}_{\text{trial}}$. An advantage of this approximation is that the correlation effects in the frustrated triangular layer is more or less taken into account in the form of the frustrated Ising chain. By contrast, spatial isotropy of the interactions in the triangular layer is lost. Note that, if one chooses the one-dimensional Hamiltonian $\mathcal{H}_{1D}^{(l)}$ as representing the interactions along the c -axis, the approximation corresponds to the type of extended mean-field approximation made by Shiba.¹⁶⁾

We have computed the r.h.s of eq. (2.2) in the following way. We first assume that the ordered state has a three-sublattice periodicity. Thus, we search for the minimum of $\mathcal{F}_{\text{trial}}$ in the space of three independent variational parameters H_A , H_B and H_C , where A , B and C denote three $\sqrt{3} \times \sqrt{3}$ sublattices of the original triangular lattice. The value of $\mathcal{F}_0$ is estimated numerically for given values of H_A , H_B and H_C via the maximum eigenvalue of the transfer matrix associated with the one-dimensional Ising Hamiltonian $\mathcal{H}_{1D}^{(l)}$. Since the interactions in $\mathcal{H}_{1D}^{(l)}$ work between up to the third-nearest neighbors, the size of the transfer matrix is 8×8 . The second term $\langle \mathcal{H} - \mathcal{H}_0 \rangle_0$ can be written as

$$\begin{aligned} \langle \mathcal{H} - \mathcal{H}_0 \rangle_0 = & -\frac{1}{3} |J_1| \{m_A m_B + m_B m_C + m_C m_A \\ & + (2R + r)(m_A^2 + m_B^2 + m_C^2) - H'_A m_A - H'_B m_B - H'_C m_C\}, \end{aligned} \quad (2.5)$$

where

$$R = \frac{J_2}{|J_1|}, \quad r = \frac{J_0}{|J_1|}, \quad (2.6)$$

$$H'_\lambda = \frac{H_\lambda}{|J_1|}, \quad (2.7)$$

$$m_\lambda = \langle \sigma_\lambda \rangle_0. \quad (2.8)$$

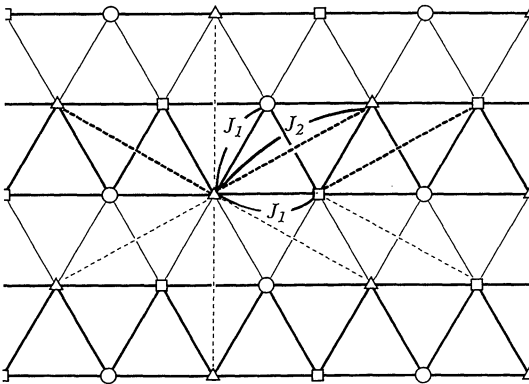


Fig. 2. The nearest-neighbor interaction J_1 and the next-nearest-neighbor interaction J_2 in the triangular layer. The interactions in the ladder, drawn by bold lines in the figure, are treated exactly as the one-dimensional Ising model. The residual interactions between the neighboring ladders, drawn by thin lines in the figure, as well as the interplane interactions, (not depicted in the figure) are treated as mean field.

Here m_λ is the sublattice magnetization per spin of the sublattice λ ($\lambda = A, B, C$), while the variational parameter H'_λ physically represents the mean-field acting on the spin on the sublattice λ . Note that m_A , m_B and m_C in eq. (2.5) are the functions of H'_A , H'_B and H'_C via eq. (2.8). In this way, we estimate the r.h.s of eq. (2.2) as a function of H'_A , H'_B and H'_C numerically. The sublattice magnetization m_λ is calculated from the differentiation of $\mathcal{F}_{\text{trial}}$ with respect to H'_λ . In our numerical calculation, differentiation is replaced by finite difference where the width in H'_λ is taken as $\Delta H'_\lambda = 10^{-6}$.

The minimization of $\mathcal{F}_{\text{trial}}$ with respect to H'_A , H'_B and H'_C has been carried out in the following manner: For given values of T , R and r , we first start from some arbitrary initial values of (H'_A, H'_B, H'_C) and calculate $\mathcal{F}_{\text{trial}}$ at this point. Then, we calculate $\mathcal{F}_{\text{trial}}$ at six neighboring points in the parameter space, $(H'_A \pm \delta H'_A, H'_B, H'_C)$, $(H'_A, H'_B \pm \delta H'_B, H'_C)$ and $(H'_A, H'_B, H'_C \pm \delta H'_C)$. If the initial point happens to have the lowest $\mathcal{F}_{\text{trial}}$ among these points, this procedure is repeated with a smaller value of $\delta H'_\lambda$. If, on the other hand, the minimum is found at one of the six neighboring points, we move to this point and repeat the above procedure around this point

with the same δH_λ value. These procedures are repeated until a stable minimum is reached. Relative error in this calculation is estimated to be about 10^{-10} in the free energy and about 10^{-5} in the sublattice magnetization. Note that, at the minimum point, the relation

$$H'_A = (4R + 2r)m_A + m_B + m_C, \text{ etc,} \quad (2.9)$$

should hold. We have checked that this relation is indeed satisfied at the obtained minimum point.

We calculate the two kinds of structure factor $F_{1/3}$ and F_1 , each corresponding to the order parameter and the uniform magnetization, defined by,

$$F_{1/3} = m_A^2 + m_B^2 + m_C^2 - m_A m_B - m_B m_C - m_C m_A, \quad (2.10)$$

$$F_1 = (m_A + m_B + m_C)^2. \quad (2.11)$$

For all types of ordered states $F_{1/3}$ takes a non-zero value, while F_1 should vanish in the partial disordered phase and takes a nonzero value in the two- and three-sublattice ferrimagnetic phases. Both F_1 and $F_{1/3}$ are directly measurable by neutron scattering measurements.

The specific heat,

$$C = -T \frac{d^2 \mathcal{F}_{\text{trial}}}{dT^2}, \quad (2.12)$$

is estimated by numerical difference, with the temperature width $\Delta T/|J_1| = 10^{-5} \sim 10^{-4}$.

§3. Results

In this section, we present the obtained results. First, let us consider the 'two-dimensional' case, $J_0=0$, studied by Mekata.⁸⁾ Depending on the value of R , qualitatively different transition behavior is observed. When the ratio $R \equiv J_2/|J_1|$ is smaller than a critical value $R_{m1} \cong 0.195$, three successive transitions occur on decreasing temperature, namely, a paramagnetic phase $\rightarrow$ a partial disordered phase $\rightarrow$ a three-sublattice ferrimagnetic phase $\rightarrow$ a two-sublattice ferrimagnetic phase. Phase transitions from one state to the other are all continuous. Temperature dependence of the specific heat and the sublattice magnetization in this range of R ($R=0.1$) is shown in Fig. 3(a) and 3(b), respectively. Successive tran-

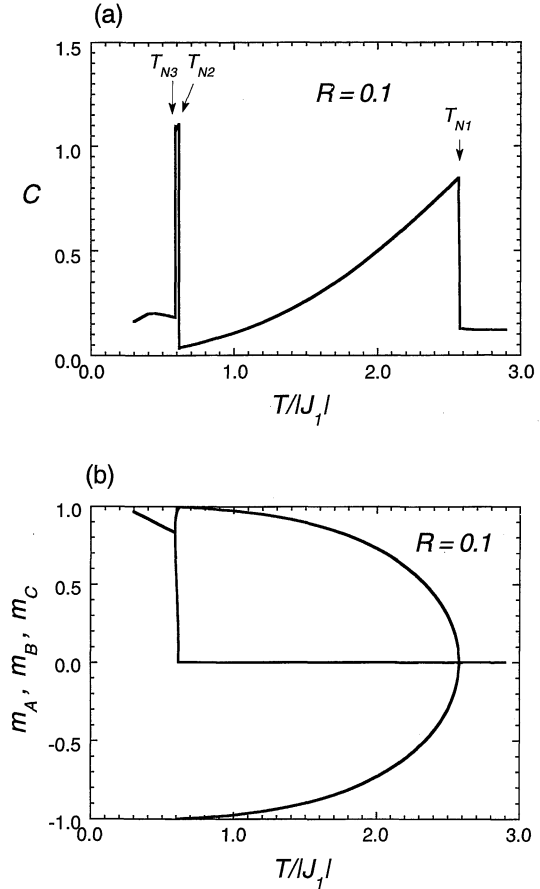


Fig. 3. Temperature dependence of the specific heat (a), and of the sublattice magnetizations (b) for $R \equiv J_2/|J_1| = 0.1$ and $J_0=0$.

sitions occur at $T_{N1}/|J_1| \cong 2.580$, $T_{N2}/|J_1| \cong 0.617$ and $T_{N3}/|J_1| \cong 0.590$. As can be seen from the figure, the partial disordered phase is stabilized over a rather wide temperature range. By contrast, T_{N2} and T_{N3} are very close to each other and the three-sublattice ferrimagnetic phase is realized only in a very narrow temperature range. In the particular case of $R=0$, the partial disordered phase is stabilized down to zero temperature with only one continuous phase transition at $T=T_{N1}$. Thus, the transition lines $T_{N2}(R)$ and $T_{N3}(R)$ go to zero in the $R \rightarrow 0$ limit.

For intermediate variable of R , $R_{m1} < R < R_{m2} \cong 0.259$, two successive phase transitions take place. The upper transition from the paramagnetic phase to the partial disor-

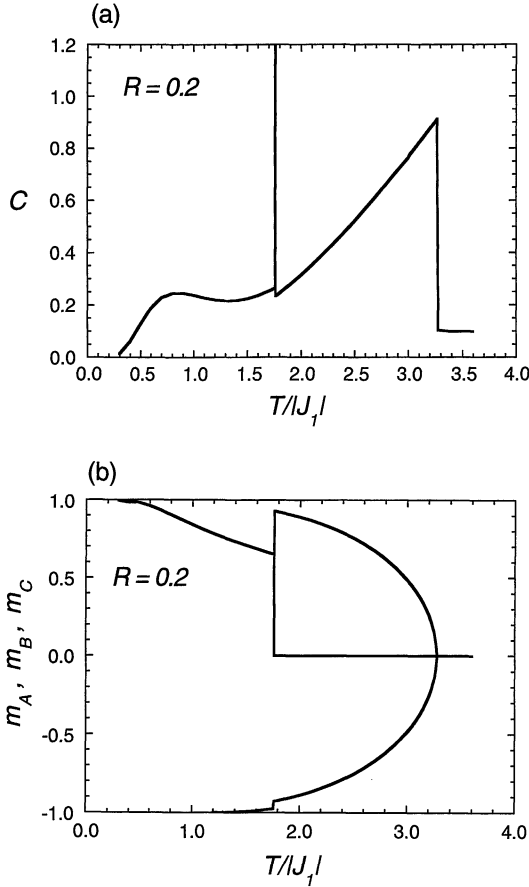


Fig. 4. Temperature dependence of the specific heat (a), and of the sublattice magnetizations (b) for $R \equiv J_2/|J_1| = 0.2$ and $J_0 = 0$. A δ -function peak at T_{N2} in Fig. (a) corresponds to the latent heat associated with a first-order transition.

dered phase is continuous while the lower transition from the partial disordered phase to the two-sublattice ferrimagnetic phase is first order. In this range of R , three-sublattice ferrimagnetic phase is not realized. The specific heat and the sublattice magnetization in this range of R ($R=0.2$) are shown in Fig. 4.

When the ratio R is larger than a critical value $R_{m2} \approx 0.259$, the system exhibits only one continuous transition from the paramagnetic phase to the two-sublattice ferrimagnetic phase. Neither the partial disordered phase nor the three-sublattice ferrimagnetic phase is realized. The specific heat and the sublattice magnetization in this range of R ($R=0.3$) are shown in Fig. 5. We note that the specific heat

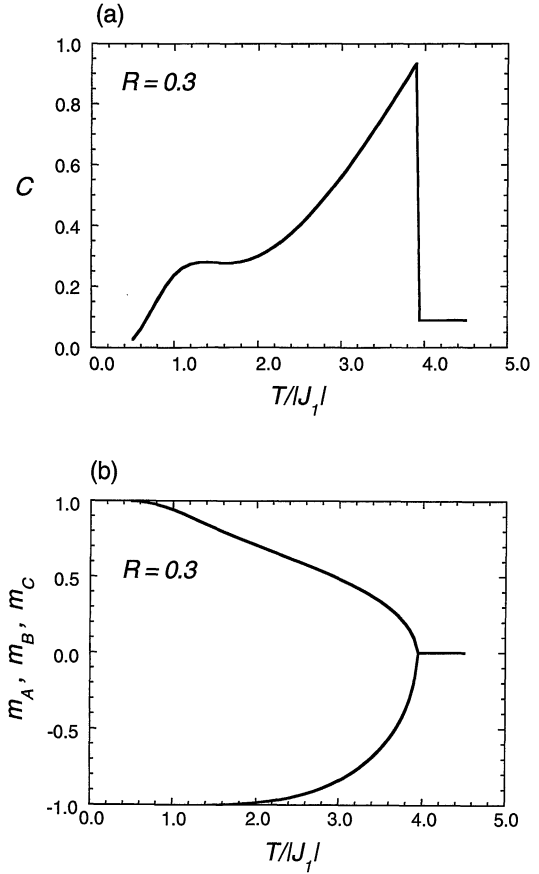


Fig. 5. Temperature dependence of the specific heat (a), and of the sublattice magnetizations (b) for $R \equiv J_2/|J_1| = 0.3$ and $J_0 = 0$.

for this range of R looks qualitatively similar to the one obtained by Monte Carlo simulations.^{12,14)}

The order parameter $F_{1/3}$ and the uniform magnetization F_1 are shown in Fig. 6 for each case of $R=0.1, 0.2$ and 0.3 . The obtained phase diagram in the $R-T$ phase is summarized in Fig. 7. There exist two multicritical points, one at $R_{m1} \approx 0.195$ and $T_{m1} \approx 1.700|J_1|$, the other at $R_{m2} \approx 0.259$ and $T_{m2} \approx 3.660|J_1|$. The former is the point where the lines of T_{N2} and T_{N3} merge, while the latter is the point where the lines of T_{N1} and the first-order transition line merge. Note that the obtained multicritical-point structure is different from that of the naive mean-field phase diagram.⁸⁾ In the mean-field approximation, only one multicritical point has been found at

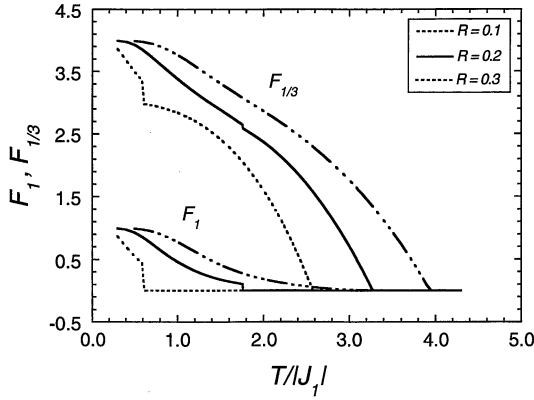


Fig. 6. Temperature dependence of the magnetic structure factors $F_{1/3}$ and F_1 , defined by eq. (2.10) and by eq. (2.11) in the text, for $J_0=0$ and $R=0.1, 0.2$, and 0.3 . F_1 vanishes in the partial disordered phase but is finite in the ferrimagnetic phases.

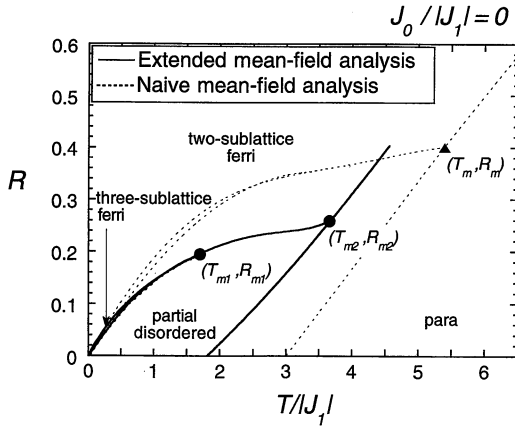


Fig. 7. Phase diagram in the $R \equiv J_0/|J_1|$ versus T plane for $J_0=0$ as obtained by the present extended mean-field approximation, which is compared with the one obtained by the naive mean-field approximation.

$R_m \approx 0.4$ and $T_m \approx 5.4 |J_1|$.

Next, we consider the effects of the interplane interaction by adding a finite $J_0 > 0$ term within the present approximation. The obtained phase diagram in the $r \equiv J_0/|J_1|$ versus $T/|J_1|$ plane is shown in Fig. 8, where we have set $J_2=0$ in Fig. 8(a) and $J_2=0.1|J_1|$ in Fig. 8(b), respectively. Again, the results are compared with those of the naive mean-field approximation. One can see from the figure that the interplane interaction J_0 works essentially in the same way as the second-neighbor inter-

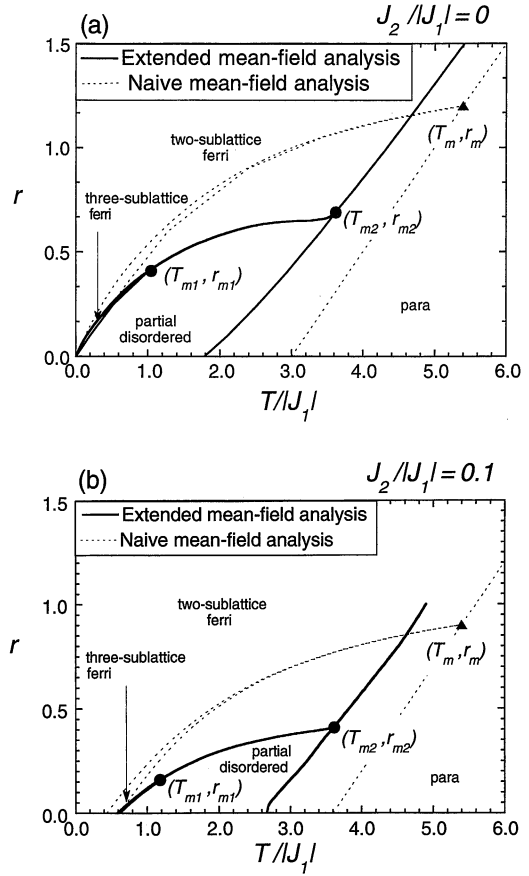


Fig. 8. Phase diagrams in the $r \equiv J_0/|J_1|$ versus T plane for $J_2=0$ (Fig. (a)) and for $J_2=0.1|J_1|$ (Fig. (b)) as obtained by the present extended mean-field approximation. The results are compared with the ones obtained by the naive mean-field approximation.

action J_2 . In particular, when the interplane interaction is sufficiently strong, $J_0/|J_1| \geq 0.69$ for $J_2=0$ and $J_0/|J_1| \geq 0.41$ for $J_2=0.1|J_1|$, the partial disordered phase is completely suppressed and the ordered phase is found to be always ferrimagnetic.

If one sets $J_0=|J_1|$ and $J_2=0$, which is the case studied by Monte Carlo simulations, our present approximation yields only one phase transition into the two-sublattice ferrimagnetic phase at $T_{N1} \approx 4.3 |J_1|$, which should be compared with the Monte Carlo results $T_{N1} \approx 2.8 |J_1|^{12)}$ and $T_{N1} \approx 2.92 |J_1|^{14)}$. By contrast, in the naive mean-field approximation, the system is found to exhibit three successive transitions, with $T_{N1} = 5.0 |J_1|$. Thus, by tak-

ing account the correlation effects in the triangular layer, our present estimate of T_{N1} comes closer to the Monte Carlo value as compared with the naive mean-field value, although it is still an overestimation.

§4. Summary and Discussion

We have studied the magnetic phase transition of the stacked-triangular Ising antiferromagnet by means of an extended mean-field approximation in which the correlation effects associated with the competing interactions in the frustrated triangular layer are taken into account. Our main purpose has been to examine the stability of the partial disordered phase in the presence of fluctuations in the triangular layer, whereas fluctuations along the c -axis are neglected. It is found that the partial disordered phase still survives within our approximation. However, the partial disordered phase turns out to be considerably suppressed relative to the ferrimagnetic phase due to the fluctuations in the frustrated triangular layer.

For example, in the case of $R=0.1$ and $J_0=0$, the partial disordered phase is found to be stabilized between $T_{N1} \cong 2.580|J_1|$ and $T_{N2} \cong 0.617|J_1|$ in the present calculation. In the naive mean-field approximation, the partial disordered phase is stabilized between $T_{N1}^{\text{MF}} \cong 3.6|J_1|$ and $T_{N2}^{\text{MF}} \cong 0.6048|J_1|$.⁸⁾ On comparison, one finds that the present T_{N1} is lower than T_{N1}^{MF} by about 29%: This has been expected, since correlation effects generally suppress the ordering. In sharp contrast to this, T_{N2} has turned out to be slightly *higher* than the corresponding mean-field value T_{N2}^{MF} , and T_{N3} is *higher* than T_{N3}^{MF} by more than 34%. This means that the correlation effects associated with fluctuations in the frustrated triangular layer favor the ferrimagnetic phase over the partial disordered phase. Likewise, the critical value of R above which the partial disordered phase ceases to exist is $R_{m2} \cong 0.259$ for $J_0=0$ in the present approximation, while in the naive mean-field approximation, the corresponding value is $R_m \cong 0.4$. Again, this suggests that the partial disordered phase is considerably suppressed by fluctuations. Similar tendency was also observed by Kaburagi *et al.* by means of a cluster variation method,¹⁰⁾

which takes account of the correlation effects associated with the frustrated triangular layer in a manner different from our present calculation. Note, however, that the cluster variation approximation is not based on the variational principle unlike the present approximation.

Our present calculation also suggests that the interplane interaction J_0 tends to enhance the ferrimagnetic phase relative to the partial disordered phase. Therefore, the existence of rather strong interplane interactions in real materials is expected to work even more unfavorably for the partial disordered phase. However, it should be noted here that in our present approximation the correlation effects associated with the *interplane* interaction along the c -axis are completely neglected. One may argue that, once the correlation effects along the c -axis were taken into account, the interplane interaction J_0 might serve to stabilize the partial disordered phase, in contrast to the results of our present approximation and of the naive mean-field approximation in which the interplane interaction is dealt with only as mean field. In order to examine this point, we have studied the relative stability of the partial disordered phase over the ferrimagnetic phase in the type of extended mean field approximation by Shiba, in which correlation effects along the c -axis are taken into account exactly as the one-dimensional Ising model whereas the intraplane interactions are dealt with as mean field.¹⁶⁾ The ratio T_{N2}/T_{N1} , which is expected to give a measure of the stability of the ferrimagnetic phase relative to the partial disordered phase, is calculated as a function of $r \equiv J_0/|J_1|$ for the case of $R \equiv J_2/|J_1|=0.1$. The result is shown in Fig. 9, together with the one of the naive mean-field approximation. As can be seen from the figure, the interplane interaction J_0 still tends to enhance the ferrimagnetic phase even when the correlation effects along the c -axis are taken into account. We thus expect that, although mean-field type approximations certainly overestimate an enhancement of the ferrimagnetic phase due to the interplane interaction, this enhancement itself is a real effect, not an artifact of our present approximation. In other words, if the partial disordered phase is less stabilized in the ‘two-dimensional’ sys-

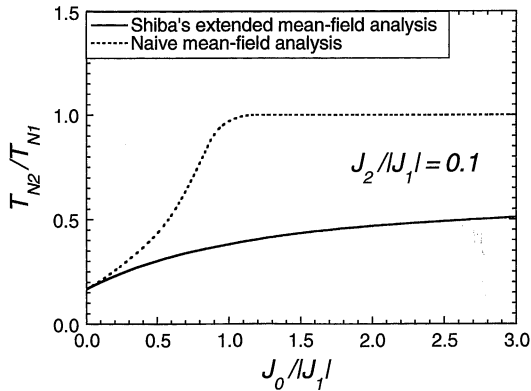


Fig. 9. The $r \equiv J_0/|J_1|$ dependence of the ratio T_{N2}/T_{N1} for the case of $R \equiv J_2/|J_1| = 0.1$, calculated by the extended mean-field approximation by Shiba. The result is compared with the one obtained by the naive mean-field approximation.

tem due to the correlation effects in the frustrated triangular layer, as indicated by our present calculation, the partial disordered phase seems to have even less chance to be stabilized in the presence of the *interplane* interaction.

Taken these observations altogether, one may conclude that the correlation effects associated with the competing interactions tend to enhance the ferrimagnetic phase and suppress the partial disordered phase in three-dimensional stacked-triangular Ising antiferromagnets. Thus, it is quite likely that the partial disordered phase is even more suppressed if one could fully take account of the correlation effects associated with the competing interactions in the triangular layer. Unfortunately, our present analysis cannot give a definitive answer whether the partial disordered phase survives as a true thermodynamic phase in a certain parameter range or not. Further study is necessary to clarify this point.

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References

- 1) M. Melamud, H. Pinto, J. Nakovsky and H. Shaked: Phys. Status Solidi **B63** (1974) 699.
- 2) W. B. Yellon, D. E. Cox and M. Einschütz: Phys. Rev. **B12** (1975) 5007.
- 3) M. Mekata and K. Adachi: J. Phys. Soc. Jpn. **44** (1977) 806.
- 4) Y. Ajiro, H. Kikuchi, T. Okita, M. Chiba, K. Adachi and M. Mekata: J. Phys. Soc. Jpn. **58** (1989) 390.
- 5) A. Farkas, D. Gaulin, Z. Tun and B. Briat: J. Appl. Phys. **69** (1991) 6167.
- 6) Y. Endoh: Butsuri (Physics) **41** (1986) 996 [in Japanese]; Y. Endoh, H. Fujimoto, H. Yoshizawa and J. P. Wicksted: private communication.
- 7) J. Wang, D. P. Belager and B. D. Gaulin: Phys. Rev. **B49** (1994) 12299.
- 8) M. Mekata: J. Phys. Soc. Jpn. **42** (1977) 76.
- 9) M. L. Plumer, A. Caillé and K. Hood: Phys. Rev. **B40** (1989) 4958.
- 10) M. Kaburagi, T. Tonegawa and J. Kanamori: J. Phys. Soc. Jpn. **51** (1982) 3857.
- 11) See, for example, S. Miyashita, H. Kitatani and Y. Kanada: J. Phys. Soc. Jpn. **B60** (1991) 1523, and references cited therein.
- 12) D. Blankschtein, M. Ma, A. N. Berker, G. S. Grest and C. M. Soukoulis: Phys. Rev. **B29** (1984) 5250.
- 13) R. R. Netz and A. N. Berker: Phys. Rev. Lett. **66** (1993) 377.
- 14) F. Matsubara and S. Ikeda: Phys. Rev. **B28** (1983) 4064; F. Matsubara and S. Inawashiro: J. Phys. Soc. Jpn. **B53** (1984) 4373; F. Matsubara and S. Inawashiro: J. Phys. Soc. Jpn. **B56** (1987) 2666.
- 15) O. Heinonen and R. G. Petschek: Phys. Rev. **B40** (1989) 9052; M. L. Plumer, A. Mailhot, R. Ducharme, A. Caillé and H. T. Diep: Phys. Rev. **B47** (1993) 14312; A. Bunker, B. D. Gaulin and C. Kallin: Phys. Rev. **B48** (1993) 15861; M. L. Plumer and A. Mailhot: to be published.
- 16) H. Shiba: Prog. Theor. Phys. **64** (1980) 466.
- 17) C. S. P. Yokoi: Phys. Rev. **B43** (1991) 8487.
- 18) K. Nakanishi: J. Phys. Soc. Jpn. **61** (1992) 2901.